TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	60.2 ± 3.25
Correlation Stability	88.2 ± 0.60
Distribution Consistency	61.7 ± 5.73
Entropy Preservation	61.3 ± 2.41
Mean Z Score Drift	0.3 ± 0.03
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	97.0 ± 1.00
Feature Accuracy	68.5 ± 2.80
Novelty Score	0.8 ± 0.02

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Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	55.979	+1.540	2.8%	OK
sex	0.680	0.860	+0.180	26.4%	Drift
cp	3.158	3.707	+0.548	17.4%	Warn
trestbps	131.690	125.802	-5.888	4.5%	OK
chol	246.693	250.306	+3.613	1.5%	OK
fbs	0.149	0.000	-0.149	100.0%	Drift
restecg	0.990	1.174	+0.183	18.5%	Warn
thalach	149.607	157.083	+7.475	5.0%	OK
exang	0.327	0.194	-0.133	40.6%	Drift
oldpeak	1.040	0.806	-0.234	22.5%	Warn
slope	1.601	1.537	-0.063	4.0%	OK
ca	0.672	0.074	-0.598	88.9%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	60.2 ± 3.25
Correlation Stability (*2)	88.2 ± 0.60
Distribution Consistency (*3)	61.7 ± 5.73
Entropy Preservation (*4)	61.3 ± 2.41
Mean Z Score Drift (*5)	0.3 ± 0.03
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	97.0 ± 1.00
Feature Accuracy (*10)	68.5 ± 2.80
Novelty Score (*11)	0.8 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.5836065573770493	0.5934426229508197	-0.004918032786885263	
F1	0.5648347697855012	0.5109475241362506	0.529969018076051	-0.034865751709449944	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5262295081967212	0.5475409836065573	0.5278688524590163	0.001639344262295072 6	
F1	0.5272786716544238	0.5271284951726232	0.5267106567192347	-0.000568014935188959 6	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196723	0.5803278688524592	0.5590163934426229	0.016393442622950817	
F1	0.5198013329305267	0.5235078311805236	0.5360918541732808	0.01629052124275404	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5360655737704919	0.5540983606557377	0.5770491803278689	0.04098360655737025	
F1	0.5231512298313296	0.5234484618875048	0.5554532514253322	0.03230202159400253	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.5819672131147542	0.5836065573770493	-0.02459016393442624	
F1	0.6070236204175147	0.5631647714444171	0.5703809280952324	-0.03664269232228219	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.5442622950819672	0.5409836065573771	-0.01311475409836066	
F1	0.4442202071436568	0.4053403033942105	0.4251509384103846	-0.01906926873327222	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.5901639344262295	0.5295081967213114	-0.02786885245901636	
F1	0.568660979990255	0.5364275496532722	0.5467861819534414	-0.02187479803681361	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5803278688524591	0.5590163934426229	0.5868852459016393	0.006557377049180313	
F1	0.5371020236217643	0.4825286896859591	0.5467638989990127	0.009661875377248252	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.5475409836065573	0.5475409836065573	0	
F1	0.3899270600110827	0.3899270600110827	0.3899270600110827	0	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5901639344262295	0.5967213114754099	0.5836065573770493	-0.006557377049180302	
F1	0.5480779520963047	0.5333646205962385	0.5299467149678682	-0.01813123712843650 6	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	59.0	1.0	4.0	125.0	222.0	0.0	2.0	124.0	1.0	3.1907690697853983
2	50.0	0.0	4.0	121.0	234.0	0.0	0.0	168.0	0.0	0.3146000644943914
3	57.0	0.0	3.0	134.0	234.0	0.0	0.0	135.0	0.0	0.2104335213796856
4	52.0	1.0	4.0	127.0	326.0	0.0	2.0	181.0	0.0	-0.1368284471156581
5	57.0	1.0	4.0	113.0	258.0	0.0	0.0	138.0	1.0	1.3491285235623798
6	39.0	1.0	3.0	115.0	215.0	0.0	0.0	173.0	0.0	-0.2305324302297392
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	51.0	1.0	3.0	144.0	239.0	0.0	0.0	191.0	0.0	0.019231183063857696
2	62.0	1.0	4.0	118.0	204.0	0.0	0.0	113.0	1.0	1.2309304365304998
3	69.0	1.0	4.0	127.0	226.0	0.0	2.0	111.0	0.0	0.8114922184891677
4	57.0	1.0	4.0	136.0	232.0	0.0	2.0	145.0	0.0	1.8316980943433687
5	36.0	1.0	2.0	119.0	226.0	0.0	0.0	153.0	0.0	-0.05407616184251364
6	50.0	1.0	4.0	140.0	311.0	0.0	2.0	166.0	0.0	1.2540421178962597
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	57.0	1.0	3.0	145.0	274.0	0.0	0.0	112.0	0.0	0.13818413751574837
2	57.0	1.0	4.0	119.0	207.0	0.0	0.0	115.0	0.0	0.6190381053541638
3	60.0	1.0	4.0	147.0	244.0	0.0	2.0	111.0	0.0	2.132682172083475
4	53.0	1.0	4.0	139.0	255.0	0.0	0.0	163.0	0.0	-0.02781255696438377
5	51.0	1.0	3.0	123.0	210.0	0.0	0.0	166.0	0.0	-0.38485417136415545
6	46.0	1.0	4.0	102.0	247.0	0.0	0.0	144.0	0.0	0.05690051223320204
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	64.0	0.0	4.0	141.0	281.0	0.0	2.0	170.0	0.0	1.0738733261054447
2	63.0	1.0	4.0	134.0	221.0	0.0	2.0	172.0	0.0	0.08456854573276734
3	59.0	1.0	4.0	124.0	255.0	0.0	0.0	176.0	0.0	-0.056613942107839405
4	55.0	1.0	4.0	122.0	204.0	0.0	0.0	171.0	0.0	-0.07199175104634992
5	61.0	1.0	4.0	133.0	275.0	0.0	2.0	108.0	1.0	3.4158418118029448
6	61.0	1.0	4.0	131.0	225.0	0.0	2.0	155.0	0.0	0.30383759718579273
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	56.0	1.0	4.0	134.0	278.0	0.0	2.0	127.0	1.0	2.650255889633906
2	67.0	1.0	4.0	132.0	222.0	0.0	2.0	156.0	0.0	0.9918855683051715
3	66.0	1.0	4.0	134.0	224.0	0.0	2.0	148.0	0.0	1.3183574984639759
4	65.0	1.0	4.0	114.0	241.0	0.0	2.0	155.0	0.0	1.606839451288536
5	54.0	1.0	4.0	130.0	226.0	0.0	2.0	113.0	0.0	1.7365962247801796
6	70.0	1.0	4.0	143.0	226.0	0.0	2.0	122.0	0.0	1.5629757773868582
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	49.0	1.0	4.0	112.0	254.0	0.0	0.0	166.0	0.0	0.06955837326239375
2	53.0	1.0	4.0	135.0	238.0	0.0	2.0	146.0	0.0	3.5521077916802835
3	49.0	1.0	2.0	119.0	239.0	0.0	0.0	160.0	0.0	-0.10824941195372031
4	63.0	1.0	3.0	121.0	206.0	0.0	0.0	131.0	0.0	0.05916522618853824
5	61.0	1.0	4.0	138.0	301.0	0.0	2.0	163.0	0.0	1.4780107512767868
6	61.0	1.0	4.0	112.0	192.0	0.0	0.0	190.0	0.0	2.512773456401679
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	59.0	1.0	4.0	120.0	321.0	0.0	2.0	143.0	0.0	1.2555035137873745
2	58.0	1.0	4.0	152.0	294.0	0.0	2.0	163.0	0.0	1.262074261816698
3	48.0	1.0	2.0	134.0	221.0	0.0	0.0	160.0	0.0	-0.3640904629353622
4	52.0	1.0	3.0	144.0	288.0	0.0	0.0	177.0	0.0	0.031112386306415468
5	57.0	0.0	4.0	124.0	364.0	0.0	2.0	148.0	0.0	2.0241386271327566
6	49.0	0.0	4.0	135.0	225.0	0.0	2.0	159.0	0.0	0.20856577897449177
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	49.0	1.0	3.0	150.0	280.0	0.0	0.0	158.0	0.0	0.13215616797138333
2	54.0	1.0	4.0	129.0	266.0	0.0	0.0	155.0	0.0	0.03150168702574858
3	49.0	1.0	4.0	122.0	219.0	0.0	0.0	171.0	1.0	0.1550314767408827
4	49.0	1.0	2.0	127.0	245.0	0.0	0.0	185.0	0.0	-0.025627611126637043
5	62.0	1.0	4.0	133.0	230.0	0.0	0.0	180.0	0.0	0.06087610819186326
6	64.0	0.0	3.0	114.0	271.0	0.0	2.0	109.0	0.0	1.0705038438903115
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	60.0	1.0	4.0	131.0	234.0	0.0	2.0	125.0	0.0	1.39411189875558
2	56.0	1.0	4.0	129.0	226.0	0.0	0.0	178.0	0.0	0.05608452951915395
3	56.0	1.0	3.0	144.0	217.0	0.0	0.0	159.0	0.0	-0.007891868725276682
4	48.0	1.0	3.0	119.0	221.0	0.0	0.0	138.0	0.0	-0.04914161874116006
5	60.0	1.0	4.0	126.0	314.0	0.0	0.0	133.0	1.0	0.11837210984386229
6	55.0	1.0	4.0	108.0	217.0	0.0	2.0	136.0	0.0	0.9098053550802458
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	61.0	1.0	3.0	121.0	283.0	0.0	0.0	153.0	0.0	0.20090674824309743
2	57.0	1.0	3.0	136.0	244.0	0.0	0.0	189.0	0.0	-0.002603947199536716
3	69.0	0.0	3.0	139.0	268.0	0.0	2.0	164.0	0.0	-0.36606557360501635
4	59.0	0.0	3.0	114.0	255.0	0.0	2.0	178.0	0.0	-0.06801013048778234
5	61.0	1.0	4.0	148.0	230.0	0.0	2.0	138.0	0.0	0.3085291762218231
6	63.0	0.0	3.0	136.0	256.0	0.0	2.0	169.0	0.0	0.09477550353869876
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6